1. Design Vehicle Hierarchy with following specifications:
- Create a base abstract class named Vehicle to hold common attributes (like registration_no, brand, and rental rate) and shared methods - calculateRent()
- Create Car and Bus class with following specifications:
1. Car (Subclass): Extends Vehicle. Adds car-specific fields like isLuxury (Boolean variable)
2. Bus (Subclass): Extends Vehicle. Adds bus-specific fields like seatingCapacity

The rent is calculated as :
1. For regular segment cars, the rent is calculated as (25 rs for first 8 hrs and 35 rs for next each hour)
2. For luxury segment cars, the rent is calculated as (40 rs for first 8 hrs and 50 rs for next each hour)
3. For buses with seatingCapacity < 12, the rent is calculated as (50 rs for first 8 hrs and 60 rs for next each hour)
4. For buses with seatingCapacity > 12, the rent is calculated as (70 rs for first 8 hrs and 80 rs for next each hour)

Design a Rental class with fields :
customer (object of Customer class with name and id)
vehicle(Object of vehicle class)

and 2 methods
1. createBooking() which records data like Customer name, vehicle number
2. generateBill() which displays the total rent in user readable format

Create RentalService class which will create objects of Renatal to create bookings and display rent for those bookings

```
// Vehicle Class

package day3_oct1.Assignement1;

public abstract class Vehicle {
     protected int regno;
     protected static int count = 0;
     protected String brand;
     protected double rentalRate;

     public Vehicle(String brand, double rentalRate) {
          this.regno = ++count;
          this.brand = brand;
          this.rentalRate = rentalRate;
     }

     public abstract double calculateRent(int hours);

     @Override
```

```
     public String toString() {
          return "Vehicle [regno=" + regno + ", brand=" + brand +
", rentalRate=" + rentalRate + "]";
     }

     public int getRegno() {
          return regno;
     }

     public void setRegno(int regno) {
          this.regno = regno;
     }

     public String getBrand() {
          return brand;
     }

     public void setBrand(String brand) {
          this.brand = brand;
     }

     public double getRentalRate() {
          return rentalRate;
     }

     public void setRentalRate(double rentalRate) {
          this.rentalRate = rentalRate;
     }
```

// Bus Class

```
package day3_oct1.Assignement1;

public class Bus extends Vehicle {
     protected int seatingCapacity;

     public Bus( String brand, double rentalRate, int
seatingCapacity) {
          super(brand,rentalRate);
          this.seatingCapacity = seatingCapacity;

     }
     @Override
     public double calculateRent(int hours) {
          double rate;
          if(this.seatingCapacity < 12) {
               rate = hours * 50;
          }
          else {
               rate = hours * 70;
          }
          return rate;
```

```
     }
     public int getSeatingCapacity() {
          return seatingCapacity;
     }
     public void setSeatingCapacity(int seatingCapacity) {
          this.seatingCapacity = seatingCapacity;
     }

     @Override
     public String toString() {
          return "Bus [seatingCapacity=" + seatingCapacity + "]";
     }
}
```

// Car Class

```
package day3_oct1.Assignement1;

public class Car extends Vehicle {
     protected boolean isLuxury;
     protected int hours;

     public Car(String brand, double rentalRate, boolean isLuxury
) {
          super(brand,rentalRate);
          this.isLuxury = isLuxury;
     }

     @Override
     public double calculateRent(int hours) {
          double rate ;

          if(this.isLuxury ) {
               if(hours < 8) {
                    rate = hours * 40;
               }
               else {
                    rate = 40 * 8; // 8 hours

                    hours -= 8;
                    rate += hours * 50;
               }
          }

          else {
               if(hours < 8) {
                    rate = hours * 25;
               }
               else {
                    rate = 25 * 8; // 8 hours

                    hours -= 8;
```

```
                    rate += hours * 35;
               }
          }

          return rate;
     }
     public boolean isLuxury() {
          return isLuxury;
     }
     public void setLuxury(boolean isLuxury) {
          this.isLuxury = isLuxury;
     }
     public int getHours() {
          return hours;
     }
     public void setHours(int hours) {
          this.hours = hours;
     }

     @Override
     public String toString() {
          return "Car [isLuxury=" + isLuxury + ", hours=" + hours
+ "]";
     }
}
```

// Customer Class

```
package day3_oct1.Assignement1;

public class Customer {
          protected int id;
          protected String name;

          protected static int count = 0;

          public Customer(String name) {
               this.id = ++count;
               this.name = name;
          }


          public int getId() {
               return id;
          }

          public void setId(int id) {
               this.id = id;
          }

          public String getName() {
               return name;
          }
```

```
          public void setName(String name) {
               this.name = name;
          }

          @Override
          public String toString() {
               return "Customer [id=" + id + ", name=" + name +
"]";
          }
// Rental Class

package day3_oct1.Assignement1;

public class Rental {
     private Customer customer;
     private Vehicle vehicle;
     private int rentalHours;

     public void createBooking(Customer customer, Vehicle vehicle,
int rentalHours) {
          this.rentalHours = rentalHours;
          this.customer = customer;
          this.vehicle = vehicle;
     }

     public void generateBill() {
          double totalRent = vehicle.calculateRent(rentalHours);
          System.out.println("customername :" +
customer.getName());
          System.out.println("vehicleReg :" + vehicle.getRegno());
          System.out.println("Totalrent :" + totalRent);

     }
}
// RentalService Class

package day3_oct1.Assignement1;

public class RentalService {

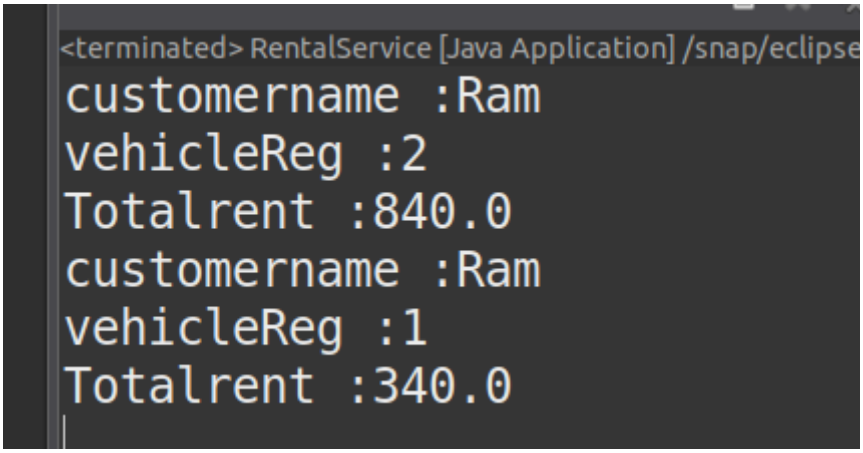
     public static void main(String[] args) {
          Customer cust = new Customer("Aman Don");
          Car c1 = new Car("Altros",34, false);
          Bus b1 = new Bus("Royal",31, 30);

          Rental r = new Rental();
          r.createBooking(cust, b1, 12);
          r.generateBill();

          r.createBooking(cust, c1, 12);
          r.generateBill();
```

```
    }
}
```

Running RentalService



2. Design Account Hierarchy with following specifications:

- The account class has account no, name and balance
- Implement withdraw and deposit methods in the class

- Create Savings and Current as subclasses of Account class

For Savings account min balance of 10000 should be maintained while withdrawal

For current account maximum of 50000 rs can be withdrawn at a time

While deposit in saving max limit is 1 lakh and for current it is 5 lakh

[To be Done in Later Assignment]